Kha Nguyen

Rust Engineer · Paris, France

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PROFILE

Rust engineer with 3+ years of production Rust, now building backend and full-stack features in TypeScript and Rust. I built the oracle engine and the DLC contract lifecycle — adaptor signatures, PSBT, Taproot/Schnorr, wallet and transaction building — and tested it against Bitcoin Core in regtest and signet. Since January 2026, I have been working on the LN Markets derivatives platform.

TECHNICAL SKILLS

Languages: Rust, Python, C++, C, TypeScript, JavaScript, SQL, HTML, CSS

Rust: rust-bitcoin, secp256k1, Tokio, Serde, Axum, Tonic, Polars, Git2, Leptos, Tauri

Bitcoin / DLC: Discreet Log Contracts (DLC), oracles, adaptor signatures, PSBT, Taproot, Schnorr signatures, Bitcoin Core (regtest, signet), Lightning Network, Eclair (ACINQ)

Python: FastAPI, Django, Pandas, PySpark, SQLAlchemy, Pytest

TypeScript: React, Angular, Nuxt, Vite, oRPC, Fumadocs, Tailwind CSS, ExpressJS, RxJS

Databases: PostgreSQL, MongoDB, MySQL, SQLite

Testing / API: cargo test, Vitest, Playwright, Pytest, Tox, Postman, Swagger

Cloud / CI/CD: AWS, Azure, Docker, Kubernetes, GitHub Actions, GitLab CI, Azure DevOps, Ansible

Observability: Prometheus, Grafana, Pyroscope

Build tools: Cargo, Make, Bazel

Ways of working: Agile / Scrum, Jira, Confluence, mdBook, Git, code review, technical documentation

Systems: Linux, macOS, Windows (WSL), Red Hat

Spoken: French, English

WORK EXPERIENCE

Ito — Paris

Mar 2025 – Present

Rust / TypeScript Engineer

Bitcoin company building DLC Markets and the LN Markets derivatives platform.

LN Markets platform — backend and full stack · Jan 2026 – Present

- Developing backend and full-stack features on the Bitcoin derivatives trading platform.
- Built REST APIs in TypeScript and Python.
- Delivered the announcement banner flow.
- Implemented trusted-device and session management.
- Built the trollbox: emoji reactions, mentions, replies and the censorship mechanism.
- Built an Eclair client for ACINQ's Lightning Network implementation.
- Wrote all the documentation pages on docs.lnmarkets.com by forking Fumadocs.
- Building the referral programme for qAssets: gold, stocks, an index and oil, margined and settled in sats.
- Built the backend with oRPC: end-to-end type-safe API contracts shared by server and client, removing a separate API schema layer.
- Built the frontend in React and TypeScript with Tailwind CSS.
- Tested with Vitest for unit tests and Playwright for end-to-end tests.

Stack · TypeScript, Rust, Python · React, oRPC, Fumadocs, Tailwind CSS, Vite · Eclair, Lightning Network · Vitest, Playwright · GitHub Actions, Docker · Linux, macOS

DLC Markets — protocol and engine in Rust · Mar 2025 – Jan 2026

- Refactored the oracle engine: event announcement, attestation and signing paths.
- Contributed to Pythia, the open-source Rust oracle for DLC trading (github.com/dlc-markets/pythia).
- Implemented the DLC contract lifecycle: offer, accept and sign message exchange, funding transaction, and Contract Execution Transactions (CETs).
- Settled contracts on-chain with adaptor signatures and oracle attestations.

- Built on the rust-bitcoin ecosystem: secp256k1, PSBT, Taproot and Schnorr signatures.
- Wrote the wallet, key management and transaction-building logic.
- Wrote end-to-end integration tests against Bitcoin Core in regtest and signet.
- Split the codebase into a cargo workspace: one crate per domain (oracle, contract lifecycle, wallet), with the domain crates free of I/O.
- Turned a PhD researcher's protocol research into production Rust code.
- Optimised and refactored existing crates; kept the workspace building on Linux and macOS.

Stack · Rust, rust-bitcoin, secp256k1, Tokio, Serde · DLC, adaptor signatures, PSBT, Taproot/Schnorr, CETs · Bitcoin Core (regtest, signet) · cargo workspace, cargo test · Linux, macOS

Allianz Trade — Paris

Sep 2023 – Mar 2025

Backend Software Engineer

Data and platform work: moving version control from Excel to Git, and building data pipelines, backend services and monitoring tools for the data science team.

Data pipelines — Python

- Extracted and analysed information from Excel and PDF documents with open-source LLMs from Hugging Face.
- Tuned a Sentence Transformers model for the data scientists.
- Built data pipelines with PySpark and Apache Kafka.
- Exposed data services through REST APIs with FastAPI.
- Built monitoring and edge-case analysis with Prometheus and Grafana.
- Implemented a rollback policy for failed transactions.

Data infrastructure and tooling — Rust

- Converted Excel tables to Parquet files with Polars for the data infrastructure.
- Profiled and optimised the existing data infrastructure with Grafana Pyroscope.
- Built CLI tools in Rust that removed manual steps from the data scientists' workflow.

Legacy modernisation — Rust and Git

- Built a linking algorithm that reconstructs a Git history graph.
- Automated the algorithm across many legacy projects.
- Optimised and refactored the existing code and made it platform-independent.
- Migrated version control from Excel to Git across the team.

Backend services — Rust and Python

- Built HTTP and gRPC services with Axum and Tonic, and Kafka consumers.
- Structured the services as a cargo workspace with one crate per domain.
- Developed REST APIs with FastAPI and implemented an event-driven architecture with Kafka.
- Deployed to on-premises infrastructure with Ansible and GitLab CI.

Stack · Rust (Polars, Git2, Axum, Tonic, Tokio, Serde) · Python (FastAPI, Pandas, PySpark, Kafka, SQLAlchemy) · PostgreSQL · Prometheus, Grafana, Pyroscope · Docker, Ansible, GitLab CI · Jira, Agile · Windows (WSL), Red Hat

French Football Federation — Paris

Sep 2021 – Sep 2023

Full-stack Developer

Club management application, online membership payment, and a user dashboard shared by several internal applications.

Front end — Angular

- Built UIs in Angular, TypeScript, HTML5 and CSS3, with responsive layouts in SCSS using Angular Material and Bootstrap.
- Prefetched data with route resolvers and protected private pages with route guards.
- Refactored class-based code into functions and moved the data model from inheritance to composition.
- Consumed REST APIs with HttpClient and RxJS, and managed state with Observables, BehaviorSubject and Subject.
- Implemented UX/UI animations.

Front end — React

- Built UIs in React and TypeScript, with responsive layouts in SCSS using MUI and Tailwind CSS.
- Consumed REST APIs with Axios and managed state with useContext.

- Added lazy loading for heavy modules.

Back end — Python / FastAPI

- Developed APIs with FastAPI and documented them with Swagger.
- Mapped the database with SQLAlchemy and optimised slow queries.
- Reduced server memory usage for file management.

Testing and delivery

- Wrote API tests with Postman, end-to-end tests with Playwright and back-end tests with Pytest.
- Ran CI/CD on Azure DevOps, containerised with Docker and opened pull requests on Azure Repos.
- Wrote documentation with mdBook and Confluence.

Stack · TypeScript, JavaScript, Python, SQL, HTML5/CSS3 · Angular, React, FastAPI, ExpressJS, RxJS · PostgreSQL, MongoDB · Playwright, Pytest, Postman · Docker, Kubernetes, Azure DevOps · Jira, Confluence, mdBook

Fintricity — England

Sep 2020 – Jun 2021

Data Engineer / Data Scientist

Weed classification: a proof of concept for a deep-learning model.

- Built the proof of concept with Jupyter Notebook, fast.ai and OpenCV.
- Compared MobileNetV3 and ResNet50 for the classification task.
- Deployed the model on AWS with Boto3 and Lambda.
- Containerised the work with Docker; tracked tasks in Jira and documented the work in Confluence.

Stack · Python, NumPy, Pandas, fast.ai, TensorFlow, Keras, PyTorch, OpenCV · AWS Lambda, Boto3 · Docker · Jira, Agile

EDUCATION

MSc Computer Science and Programming — Sep 2017 – Feb 2018
Loughborough University — Loughborough, England

MEng Mechatronics Engineering — Sep 2014 – Sep 2019
University of Technology of Compiègne (UTC) — Compiègne, France

ACADEMIC EXPERIENCE

3rd place — Créathon UTC — Oct 2019
Compiègne, France — algae-based CO2 filter for cars, to cut emissions and produce biomass.

Robotics Engineer Intern — Trimble — Feb 2019 – Aug 2019
Nantes, France — camera and inertial sensor synchronisation in C++.

- Interpolated RealSense D345 camera and DMU11 IMU data with SLERP.
- Published the fused stream as a ROS topic.

Special Prize — UTC Hackathon — Oct 2018
Compiègne, France — LED animated clock driven by an Arduino Uno.

Process Engineer Intern — Valeo — Feb 2018 – Aug 2018
Reims, France — radiator crimping inspection with four cameras.

- Designed machine accessories in CATIA V5.
- Wrote the user guide and the safety documentation for the inspection machine.

UTC Hackathon — smart refrigerator — Oct 2016
Compiègne, France — program that helps a bartender arrange beer bottles in a refrigerator.